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Total Cost of Care associated with Opioid Use Disorder Treatment

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Abstract

The opioid epidemic in the United States disproportionately affects Medicaid beneficiaries than other groups. This results in a significant financial burden on state Medicaid programs. In this analysis, we investigate the association of medication for opioid use disorder (MOUD) treatment initiation and linkage to ongoing care on overall healthcare costs of Medicaid Fee-for-Service patients. We conducted a retrospective study among adult patients diagnosed with opioid use disorder (OUD) and who had a clinical encounter at a safety-net institution in Denver Colorado in 2020. Three categories of MOUD status of patients were defined: 1) identified with OUD but did not receive MOUD; 2) initiated MOUD but not linked to ongoing treatment and 3) received MOUD and linked to ongoing treatment. Our outcome variable was per-member per-month total healthcare cost. We estimated a multivariable model to test the association between healthcare cost and MOUD status, while controlling for demographic and risk classification variables. We found that in individuals with OUD who initiated MOUD treatment but were not linked to ongoing care had the highest healthcare cost, while those who were linked to ongoing MOUD treatment had the lowest healthcare cost. MOUD treatment is not only effective at addressing the significant morbidity and mortality burden of OUD but also associated with decreased financial cost, which is disproportionately incurred by Medicaid. Additional policy and care delivery changes are needed to focus efforts to improve linkage to ongoing treatment.

Keywords

Medication for Opioid Use Disorder; Healthcare Cost; Medicaid

1. Introduction

The opioid epidemic in the United States (US) has steadily increased since the late 1990s, with an estimated 2.7 million individuals diagnosed with opioid use disorder (OUD)

(SAMHSA, 2021) and over 70,000 deaths due to related drug overdose (National Institute on Drug Abuse, 2022) in 2020. In addition to alarming effects on morbidity and mortality, the opioid epidemic has also resulted in high financial costs to society. The cost of OUD and related overdose was estimated to be \$1.02 trillion in 2017 in the US (Florence et al., 2021).

Medicaid beneficiaries have higher rates of mental and substance use disorders (McAdam-Marx et al., 2010) and are at a greater risk for OUD as compared to the general population (Adelmann, 2003). As a result, OUD has been documented to impose significant financial burden on state Medicaid programs (Leslie et al., 2019). Additionally, it has been found that non-OUD related healthcare use has been a major driver of healthcare costs for Medicaid patients with OUD (Leslie et al., 2019). The total OUD-related cost to Medicaid, including costs both directly related to OUD treatment and indirectly related to poor health outcomes associated with OUD, was estimated to be \$8.42 billion in 2013 or 3.2% of total Medicaid costs (Leslie et al., 2019).

Although the impact of OUD on individuals and society is substantial, treatment is effective. The OUD care continuum provides a framework for evaluating treatment (Prieto et al., 2019; Williams et al., 2019). It is composed of sequential stages that include OUD diagnosis, initiation of treatment, linkage to ongoing care and retention in care. Treatment includes medications for OUD (MOUD) and has been associated with favorable health outcomes (Williams et al., 2019). MOUD has been demonstrated to have superior effectiveness than behavioral treatment alone (Connery, 2015) as MOUD recipients have a lower risk of mortality (Larochelle et al., 2018) and theoretical models have also established its cost effectiveness (Fairley et al., 2021). This care continuum has identified linkage to ongoing treatment as an important step in patient engagement (Prieto et al., 2019). However, it is not yet known how individuals progressing through these early stages of the OUD care continuum differ in terms of their healthcare costs, and examination of the potential cost savings of MOUD is needed to inform policy and care delivery. In this analysis, we investigate the association of MOUD treatment initiation and linkage to ongoing care on overall healthcare costs of Medicaid Fee-for-Service patients.

2. Materials and Methods

We conducted a retrospective study among adult (18 years and older) patients diagnosed with OUD at Denver Health, a safety-net institution in Denver Colorado in 2020. Denver Health includes a Center for Addiction Medicine, which coordinates as a hub and spoke model of care (Brooklyn and Sigmon, 2017). The specialty hub includes an Opioid Treatment Program, office-based opioid therapy, and specialty psychiatric care. Other parts of the integrated safety-net system, including the hospital, emergency department, county jail, federally qualified health centers, serve as spokes. Regardless of where persons with OUD enter the system, they can be identified, treated, and linked to the right level of ongoing treatment. The electronic health record was used to identify patients with evidence of OUD in the year 2020. Evidence of OUD included International Classification of Diseases (ICD-10) diagnoses for OUD, opioid misuse and opioid poisoning, Clinical Opiate Withdrawal Scale assessment (Wesson and Ling, 2003), positive lab result for heroin, or

medication order for buprenorphine or methadone. We provide details of proportion of enrollees identified with OUD using each of these methods in Supplemental Table 1.

Our outcome variable was per-member per-month (PMPM) total healthcare cost during 2020. To ensure access to complete data, patients in the current analysis were restricted to those who were eligible for Medicaid fee-for-service (FFS) coverage during the year 2020 and attributed to Denver Health for primary care through the Colorado State Medicaid Accountable Care Organization (ACO). In Colorado, the majority of Medicaid members are enrolled in FFS coverage.

The primary independent variable was a three-category variable that described the MOUD (Buprenorphine or Methadone) status of patients based on initial stages of the OUD care continuum: 1) identified with OUD but did not receive MOUD; 2) initiated MOUD but not linked to ongoing treatment; and 3) received MOUD and linked to ongoing treatment. We defined a patient to be linked to ongoing treatment as someone who received at least one MOUD dose in an outpatient area at Denver Health able to provide ongoing, longitudinal MOUD care. This was informed by the HIV care continuum (Gardner et al., 2011) and aligned with the US Health Resources and Services Administration HIV/AIDS Bureau systems-level performance measure of “Linkage to HIV Medical Care” based on attending at least one routine HIV medical care visit (HRSA 2022). This conceptualization of the OUD care continuum differs from other models in order to examine differences in MOUD treatment that includes expanded access points such as inpatient, emergency services and correctional care settings that are not designed to support ongoing longitudinal care. We also included sociodemographic information, including age in years, sex (female or male), race/ethnicity (African American/Black, American Indian/Alaskan Native, Asian, Hispanic, White and Other/Unknown) and clinical risk group (CRG) classification representing core health status and predicting future healthcare utilization and cost (3M Health Information Systems, 2022). Denver Health calculates CRG based on 18-months of historical data (July 1, 2019 – December 31, 2020 for the study period) and utilizes CRG for tiered risk stratification, aggregating nine categories of CRG into 3 groups (1–2, 3–5, and 6–9) corresponding to increasing severity and number of medical conditions (Johnson et al., 2015).

Descriptive statistics were used to characterize sociodemographic and clinical characteristics of the cohort overall and by MOUD status; differences in distribution of characteristics were tested using Pearson’s chi-square test for categorical variables and a non-parametric k-sample median test for continuous age. A multivariable generalized linear model (GLM), with log link and gamma distribution to account for skewed cost data (Deb and Norton, 2018; Diehr et al., 1999) was applied to empirically test the association between PMPM healthcare cost and MOUD status, while controlling for sociodemographic and clinical covariates. We confirmed our choice of log link using the Box-Cox approach test and gamma distribution using the modified Park test (results of these tests are included in the Supplement). For the GLM, due to sample sizes, we collapsed the race/ethnicity categories as Non-Hispanic Black, Non-Hispanic White, Hispanic and Other. Finally, using the estimated model, we calculated the average marginal effects of MOUD status on PMPM cost and used analysis of variance contrast tests to estimate the differences in predicted

average PMPM costs and 95% confidence intervals (CI) between MOUD status groups using the category of OUD identified but not receiving MOUD as the reference group.

Select sensitivity analyses were performed. We attempted to control for large variation in PMPM costs across our sample by Winsorizing PMPM costs (at 2% for low outlier costs and 98% for high outlier costs) and used these costs as dependent variable for the GLM. Additionally, to ensure our results were robust to different time periods within 2020, we estimated the above-mentioned specification of GLM in different strata based on the calendar quarter of the first date of MOUD receipt in 2020. Also, since 55% of patients linked to MOUD treatment in our sample had their first 2020 MOUD receipt in January and may have been continuing treatment from the prior year, we also specified the GLM with these individuals excluded. All analyses were conducted using Stata 16 (StataCorp, 2019) and utilized a significance level of $\alpha=0.05$. This study was approved by the Colorado Multiple Institutional Review Board.

3. Results

During 2020, there were a total of 54,388 Medicaid FFS attributed patients: 31,570 were aged ≥ 18 years and 905 were identified with OUD. After checking for missingness in the cost and CRG data, 850 Medicaid FFS patients who were diagnosed with OUD were included in our study. Of these, 48.2% ($n=410$) did not initiate MOUD, 6.7% ($n=57$) initiated MOUD but were not linked to ongoing care, and 45.1% ($n=383$) initiated and were linked to ongoing MOUD (Table 1). We did not observe any significant differences in age, sex, or race/ethnicity across the three groups. All three MOUD groups were skewed toward the significant chronic disease category with the fewest participants in the lowest CRG group.

Based on the covariate adjusted GLM, we found that average marginal estimates of PMPM healthcare cost in 2020 were \$1,425 (95% CI \$1117 – \$1733) for those who did not initiate MOUD treatment, \$2,385 (95% CI \$1668 – \$3103) for those who initiated MOUD treatment but were not linked to ongoing care, and \$990 (95% CI \$813 – \$1168) for those who initiated and were linked to ongoing MOUD treatment (Figure 1). Differences in predicted average PMPM healthcare costs between the MOUD status groups, using cost of patients who did not initiate MOUD as the reference group are shown in Supplemental Figure 1. Those that initiated MOUD but were not effectively linked to care had highest costs, compared to both those who did not initiate MOUD and those who were linked to ongoing MOUD treatment. Findings from our sensitivity analyses were mostly similar and supported results from our primary model (Supplemental Figures 2 – 4). While we observed an increase in MOUD linked from \$990 to \$1236 when the January group was excluded, it was still lower than the MOUD not linked group and followed the trend we observed in our primary model.

4. Discussion

This study found that individuals with OUD who initiated MOUD treatment but were not linked to ongoing care had the highest PMPM healthcare cost, while those who were

linked to ongoing MOUD treatment had the lowest PMPM healthcare cost. As expected, this is consistent with evidence in different healthcare settings reporting an association between medication compliance and lower healthcare costs and use (Roebuck et al., 2011). It is interesting to note that while we only considered the linkage to ongoing medications specifically related to OUD, this was associated with reduced overall healthcare costs. Efforts to encourage and promote linkage to ongoing treatment may be an important factor in moderating financial burden on public healthcare resources.

Our findings also revealed that individuals who initiate treatment, but are not successfully linked to ongoing treatment, pose the highest financial burden to Medicaid. Although our analysis did not control for individual behavior, these individuals might be risk averse to medical treatment or mistrust the healthcare system. These behavioral characteristics may result in an increase utilization of healthcare services increasing overall cost of care (Weiden et al., 2004). Further research is required to better assess the reasons that result in non-linkage to treatment. This group should be a focus of future public health interventions related to OUD.

This analysis adds to the ongoing effort to ensure Medicaid coverage of MOUD treatment. The SUPPORT Act legislation of 2020 requires state Medicaid to cover MOUD treatment including related counseling and behavioral therapy (Centers for Medicare and Medicaid Services, 2020). Additional policy changes are necessary to facilitate access to OUD treatment for all persons. A Kaiser Family Foundation analysis of the 2017 National Survey on Drug Use and Health reported that only 44% of Medicaid nonelderly adults with OUD utilized treatment in the past year and only 34% across all insurance statuses (Orgera and Tolbert, 2019). Our analysis supports additional research into coverage for services and resources linking persons with OUD to ongoing treatment, potentially through traditionally non-covered extended healthcare team members such as certified (non-licensed) addiction counselors.

It is important to note that this analysis of costs associated with healthcare utilization was conducted during the first year of the COVID-19 epidemic, when there were major changes in the provision and access to care, including a need to avoid large volumes of patients and fewer in-person visits facilitated by changes to regulations for MOUD prescriptions and a shift to telemedicine for MOUD monitoring (Jacka et al., 2021). Several studies report minimal impact of the provision of MOUD care due to these adaptations (Alexander et al., 2021); yet other studies have reported more emergency room use among frequent users with substance use disorder (Chandran et al., 2021), and higher risk of overdose (Linas et al., 2021), which could inflate the proportion of persons with OUD presenting for emergency care. It is therefore possible that the differential costs we observed associated with different levels of engagement in MOUD for 2020 may not be generalizable to healthcare costs in non-pandemic time periods. Despite the potential impact of the COVID-19 epidemic, our sensitivity analysis stratifying by 2020 calendar year quarters did not reveal any differences and the January through March time frame would have mainly represented a pre-COVID-19 time period.

Additional limitations of our analysis include the use of aggregated Medicaid FFS total cost of care data for individuals who are attributed to a single institution. Our results may not be generalizable in other settings and contexts such as rural settings or those with different access to MOUD treatment. We only included linkage to care within the Denver Health system, and therefore could have underestimated the patients who were effectively linked to care. We could only access aggregated cost data due to the current limitation to claim level data related to the confidentiality of substance use disorder (SUD) patient records under US Code of Federal Regulations 42 CFR Part 2 (SAMHSA 2022). This regulation prohibits the disclosure of any information that would identify a person as having a SUD unless the person provides written consent. The CARES Act of 2020 updated 42 CFR Part 2 to align with the Health Insurance Portability and Accountability Act of 1996 (HIPAA) Privacy Rule to allow SUD-related disclosure for the purposes of treatment, payment, and health care operations. However, current regulations remain in effect until new regulations are promulgated. Therefore, claims that contain SUD information are not shared and only member-level aggregated total cost of care data was available for this analysis. This did not allow more refined analyses into types of costs, limiting costs only after initiation of MOUD or leveraging claims data to identify OUD diagnosis or care continuum progression. We also were unable to account for the severity and duration of patients' OUD which might impact their initiation and linkage to ongoing MOUD treatment as well as costs. This limitation was partially addressed by the sensitivity analysis excluding patients potentially continuing treatment from the prior year which produced similar results to our full model. Finally, our analysis only addressed initiation of MOUD and linkage to ongoing care but did not address subsequent retention stages of the OUD care continuum.

5. Conclusion

MOUD treatment is effective at addressing the significant morbidity and mortality burden of OUD as well as associated with decreased financial cost, all of which are disproportionately incurred by the Medicaid population. Individuals who initiated and were linked to ongoing MOUD treatment had the lowest total healthcare cost, and those who initiated treatment but were not linked to ongoing care had higher total healthcare costs than those who did not initiate treatment. Additional efforts should be made to characterize these populations, including the sources of these cost differences as well as barriers to linkage and retention.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Disclosure of Funding and Conflicts of Interest

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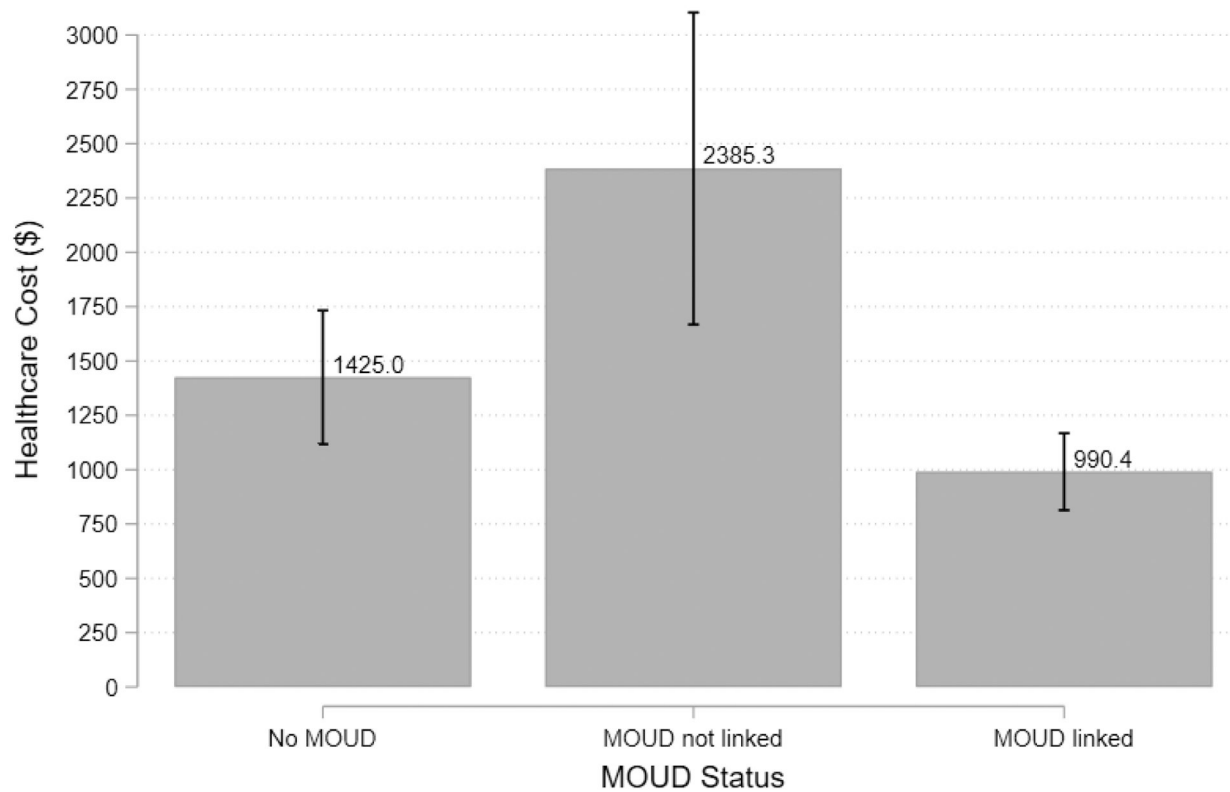


Figure 1.

Marginal^a estimates of average per-member per-month (PMPM) healthcare costs based on Medication for Opioid Use Disorder (MOUD) status^b groups for Medicaid Fee-for-service enrollees who were identified with opioid use disorder (OUD) at Denver Health (DH) during 2020. Marginal estimates are displayed with their 95% confidence intervals.

Notes: Data sources – Medicaid FFS claims and electronic health records at Denver.

^a Predictions based on Multivariable Generalized Linear Model (GLM) with log link and gamma distribution. This model controls for member's age, sex, race/ethnicity and CRG health status. Using the estimated model, we calculated the average marginal effects of MOUD status on PMPM cost.

^b MOUD Status is categorized as follows: No MOUD = Identified with opioid use disorder but did not receive MOUD, MOUD not linked = Received MOUD but not linked to ongoing treatment, and MOUD linked = Received MOUD and linked to ongoing treatment. Linked to ongoing treatment defined as someone who received dose/s at an outpatient area at DH

Table 1

Characteristics of Medicaid Fee-for-service enrollees identified with opioid use disorder (OUD) at Denver Health (DH) between January 1-December 31,2020, by Medication for OUD (MOUD) Status^a. Values are expressed as counts (n) and % unless otherwise specified.

Characteristic	All 850 (100)	MOUD Status			p-value ^b
		No MOUD 410 (48.2)	MOUD not linked 57 (6.7)	MOUD linked 383 (45.1)	
Age (years), median [(interquartile range (IQR))]	41(32,54)	45 (33,57)	38 (30,52)	39 (32,50)	0.456
Sex					0.316
Female	401 (47.2)	183 (44.6)	30 (52.6)	188 (49.1)	
Male	449 (52.8)	227 (55.4)	27 (47.4)	195 (50.9)	
Race/Ethnicity ^c					0.094
African American/Black	83 (9.8)	50 (12.2)	3 (5.3)	30 (7.8)	
American Indian/Alaskan Native	9 (1.1)	8 (2.0)	0 (0)	1 (0.3)	
Asian	5 (0.6)	3 (0.73)	0 (0)	2 (0.5)	
Hispanic	318 (37.4)	153 (37.3)	17 (29.8)	148 (38.6)	
White	417 (49.1)	187 (45.6)	36 (63.2)	194 (50.7)	
Other/Unknown	18 (2.1)	9 (2.2)	1 (1.8)	8 (2.1)	
CRG Categories ^d					0.001
1–2	109 (12.8)	68 (16.6)	12 (21.1)	29 (7.8)	
3–5	326 (38.4)	150 (36.6)	18 (31.6)	158 (41.3)	
6–9	415 (48.8)	192 (46.8)	27 (47.4)	196 (51.2)	

Notes: Data source – Electronic health records at Denver Health.

^aMOUD Status is categorized as follows: No MOUD = Identified with opioid use disorder but did not receive MOUD, MOUD not linked = Received MOUD but not linked to ongoing treatment, and MOUD linked = Received MOUD and linked to ongoing treatment. Linked to ongoing treatment defined as someone who received dose/s at an outpatient area at DH

^bp-values based on Pearson's Chi-Square test for categorical variables and Median test (Stata command=*median*) for comparison of medians.

^cRacial/ethnic categories other than Hispanic are non-Hispanic.

^dCRG Core Health Status ranges in values from 1 to 9.